

1 Human-Trace Source Alignment

CASAS
Home Records



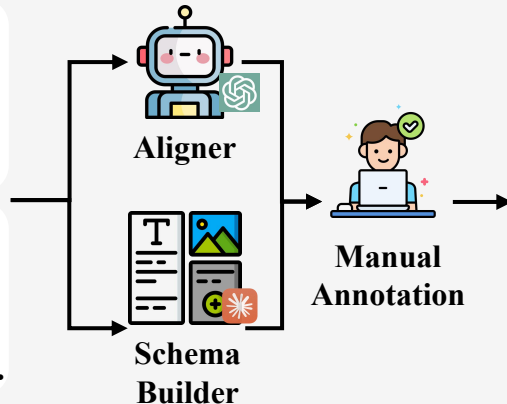
ARAS
Resident Activity



HD-EPIC
Egocentric Interaction



HOMER+
Simulated routine evidence



Unified Event Schema

Time	Actor	Object
Activity	Source state	
Destination state	...	

Grounding & Normalization

1. temporal alignment
2. activity-object linkage
3. evidence provenance

2 P4D-Bench Construction



Unified Event Schema

World Generation Library



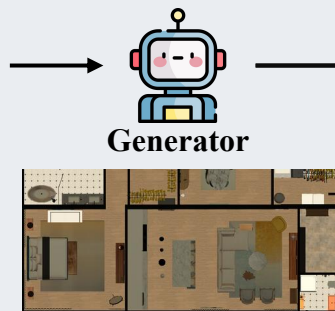
Household profile



World-transition rule



Task templates



Generator

Executable household



Worlds

Static world
Routine world
Random world



Evaluation Tasks

1. N1-N5 Tasks
2. VLN
3. EQA



P4D Bench